

# SATYAM PRIYADARSHI

🔗 satyamp03 | 🌐 Satyam Priyadarshi | 🆔 ORCID

## RESEARCH PROFILE

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Research technician at the de Boer Lab (UBC School of Biomedical Engineering), building open-source benchmarking infrastructure for genomic language models (GAME; Luthra *et al.*, 2025) and working hands-on with state-of-the-art seq2func models including AlphaGenome, Borzoi, ChromBPNet, and DREAM-RNN. Currently extending into the modelling side through generative sequence modelling with diffusion and flow-matching-based methods. Background in physics and astronomy, with an undergraduate honours thesis at the Belle II group at UBC.

## EDUCATION

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**Bachelor of Science**, University of British Columbia  
Combined Honours in Physics and Astronomy

📅 2018 – 2023

• **Honours Thesis:** *Search for Low-Mass Dark Photons at Belle II*

Supervised by Dr. Christopher Hearty, Professor & IPP Scientist

• **Co-op Research:** Quantum Materials Design Lab, SBQMI

Supervised by Dr. Meigan Aronson, Dean of Science (2018–23) & PI, and  
Dr. Alannah Hallas, Associate Professor & PI

## PUBLICATION

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1. Ishika Luthra, **Satyam Priyadarshi**, Rui Guo, Lukas Mahieu, Niklas Kempynck, Damion Dooley, Dmitry Penzar, Ilya Vorontsov, Yilun Sheng, Xinming Tu, Adam Klie, Shiron Drusinsky, Alexander Floren, Ethan Armand, Kaur Alasoo, Georg Seelig, Ryan Tewhey, Peter Koo, Vikram Agarwal, Sager Gosai, Luca Pinello, Michael A. White, Avantika Lal, Julia Zeitlinger, Katherine S. Pollard, Maxwell Libbrecht, Hannah Carter, Sara Mostafavi, Ivan Kulakovskiy, Will Hsiao, Stein Aerts, Jian Zhou, and Carl G. de Boer. “GAME: Genomic API for Model Evaluation”. Preprint 2025 [*bioRxiv*]

## ABSTRACT AND POSTER PRESENTATION

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1. Luthra, Ishika, **Priyadarshi, Satyam**, et al. “GAME: Genomic API for Model Evaluation”. May 2026. CHSL 90th Symposium – AI in Biology (Cold Spring Harbor, NY) Poster Presentation
2. Luthra, Ishika, **Priyadarshi, Satyam**, et al. “GAME: Genomic API for Model Evaluation”. June 2025. Cascadia Advanced Genomics Conference (Vancouver, BC) Poster Presentation
3. William de Jong, Grant, **Priyadarshi, Satyam**, and Adams, Keith. L. “Subgenome-Dominant Expression and Alternative Splicing in Response to Sclerotinia Infection in Polyploid *Brassica napus* and Progenitors”. July 2024. ICAR 2024 (San Diego, CA) Poster Presentation
4. William de Jong, Grant, **Priyadarshi, Satyam**, and Adams, Keith. L. “Subgenome-Dominant Expression and Alternative Splicing in Response to Sclerotinia Infection in Polyploid *Brassica napus* and Progenitors”. January 2024. PAG 31 (San Diego, CA) Abstract Presentation
5. Li, Xiyang, Oudah, Mohamed, Reig-I-Plessis, Dalmau, **Priyadarshi, Satyam**, et al. “Spin dynamics in a metallic spin chain  $\text{Ti}_4\text{MnBi}_2$ ”. March 2023. APS March Meeting 2023 (Las Vegas, NV) Abstract Presentation

## RESEARCH EXPERIENCE

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- **Research Assistant & Lab Technician (de Boer Lab)**  
*School of Biomedical Engineering @ UBC (Full Time)*

📅 October 2024 – Present  
*Vancouver*

  - **Generative sequence modelling (in progress):** Implementing **diffusion & flow-matching**-based approaches for single-nucleotide-resolution genomic modelling.
  - **Developed the open-source Genomic API for Model Evaluation (GAME) framework:** Designed **client–server REST APIs** that decouple genomic deep-learning models (*Predictors*) from benchmark datasets (*Evaluators*), enabling **standardized, reproducible benchmarking** of sequence-to-function (seq2func) models across independent compute environments.
  - **Engineered a load orchestrator:** Built a transparent “scatter–gather” orchestration service (**FastAPI**, **asyncio**) that partitions large sequence batches, dispatches them across multiple HPC nodes in parallel, and reassembles predictions — with automatic msgpack/JSON content negotiation and a **distributed consistency check** on returned results.
  - **Built the LLM-powered Matcher module:** Integrated a locally-hosted LLM (**Gemma 3** via **Ollama**) to automatically reconcile free-text task, cell-type, & species labels between heterogeneous models and benchmarks, using a **tournament-style elimination** strategy to resolve label sets exceeding the model’s context window.
  - **Containerized & deployed genomic models:** Packaged state-of-the-art models (e.g., **AlphaGenome**, **ChromBPNet**, **Borzoï**, **DREAM-RNN**) as portable **Apptainer/Conda** containers, resolving conflicting CUDA & dependency stacks to enable reproducible, GPU-accelerated evaluation on shared HPC infrastructure.
  - **Ran large-scale evaluation workloads:** Executed GPU benchmarking & analysis pipelines on **UBC ARC Sockeye & Digital Research Alliance of Canada** clusters, managing SLURM job submission, scheduling, & multi-node resource allocation.
  - **Designed a fault-tolerant error protocol:** Implemented structured, typed error reporting with graceful HTTP failure handling, so distributed evaluation runs fail diagnosably rather than silently across nodes.
  - **Authored developer documentation & API schema:** Wrote the **API schema reference** & module-building guides that enable external labs to implement compliant GAME modules, lowering the barrier to community contribution to the benchmarking ecosystem.
- **Research Assistant & Lab Technician (Adams’ Lab)**  
*Beaty Biodiversity Research Centre @ UBC (Full Time)*

📅 August 2023 – August 2024  
*Vancouver*

  - **Developed computational scRNA-seq pipelines:** Built end-to-end analysis workflows for single-cell RNA-seq data, quantifying gene expression heterogeneity at single-cell resolution to extend the lab’s bulk RNA-seq capabilities.
  - **Engineered transcriptomic workflows:** Built robust pipelines for **Illumina RNA-seq data**, performing quality control, alignment (**STAR**, **HISAT2**), and differential expression analysis (**DESeq2**, **edgeR**) to isolate significant gene candidates.
  - **Investigated gene regulation mechanisms:** Implemented specialized analysis using **rMATS** & **SUPPA2** to detect and quantify **alternative splicing events**, providing deeper insights into transcript isoform diversity.
- **Undergraduate Research Assistant (Aronson-Hallas’ Lab)**  
*Stewart Blusson Quantum Matter Institute (Co-op & Work-Learn)*

📅 February 2020 – May 2023  
*Vancouver*

  - **Improved material growth efficiency:** Increased efficiency for **flux & solid-state growths**, analyzing single crystals & poly-crystalline powder materials using **Neutron Scattering**, **X-Ray**

**Diffraction, Scanning Electron & Scanning Tunneling Microscopy (SEM/ STM/ TEM), Energy Dispersive X-Ray (EDX) Spectroscopy, Physical & Magnetic Property Measurement Systems.**

- **Investigated new materials:** Researched over 10 topological materials & compounds, characterizing **frustrated magnetic systems, superconductors, & Weyl semimetals.**
- **Project leadership for engineering efforts:** Led development of a **new ternary phase for superconductive & antiferromagnetic binaries**, co-authoring a paper on spin dynamics in spin-chain systems.
- **Enhanced data management:** Organized and automated 8 years of sample growth data using **Pandas and SQL** for easy retrieval and analysis.
- **Conducted data analysis:** Analyzed XRD data using **Pawley and Rietveld refinements** to **determine lattice parameters** of crystals, identify **space groups**, and ensure **accurate characterization**. Additionally, performed analysis using Energy Dispersive Spectroscopy and analyzed large-scale PPMS and MPMS data.
- **Mentorship:** Mentored graduate students, post-docs, and fellow undergraduates overseeing their sample growth and providing guidance on lab equipment usage while prioritizing safety protocols.

- **Undergraduate Honours Thesis Researcher** 📅 September 2022 – April 2023  
*Belle II Research Group @ UBC — Belle II Experiment @ SuperKEKB* Vancouver
  - **Processed large-scale simulation data:** Reconstructed and analyzed over **20 million simulated Monte Carlo events**, applying particle tracking and clustering algorithms to identify potential dark photon candidates.
  - **Developed signal selection criteria:** Investigated and tested the performance of various kinematic cuts to optimize the selection of events consistent with the  $e^+e^-\gamma$  **signature**, forming the core of the thesis analysis.
  - **Validated analysis against varied conditions:** Analyzed **Bhabha and digamma background simulations** from four different experimental runs to validate the analysis strategy against changing beam conditions and reduce statistical uncertainties.

## HONOURS AND AWARDS

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- UBC Staff Honorarium – Work “Above and Beyond” (September 2023)
- Dean’s Honour List (April 2020)
- Outstanding International Student Award (September 2018)

## CERTIFICATION

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- **Google Data Analytics Professional Certificate** 📅 June 2024  
*Data Cleaning, Preparation, Statistical Methods, ML Analysis, Visualization, Ethics* Coursera

## RELEVANT PROJECTS

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- **Developed an AI-powered recommendation engine:** Implemented **collaborative filtering** for personalized product recommendations based on user history. Pre-processed **250+ TV shows** data, developed the model, and deployed a **Flask API**. Evaluated performance using **precision and recall** metrics. (July 2024)

- **ASA DataFest UBC**: Implemented **ridge regression** in a **content-based filtering recommender system** for pro-bono attorneys, **enhancing matching accuracy and system efficiency**. Streamlined attorney-client matching through **feature selection**, **multicollinearity** handling, and regularization. **Improved robustness** to noise and irrelevant features, optimizing performance and client satisfaction. (April 2023)
- **iQuHACK 2023 – IonQ Challenge**: Developed a **quantum algorithm** that **encoded camera-captured images into quantum circuits** to create an **image classifier**. Utilized **FRQI representation** to efficiently encode & **process image data on a quantum computer**, enhancing accuracy. (January 2023)
- **NASA Space Apps Challenge 2022**: Used **ArcGIS Pro** & **data mining** to **preprocess & analyze netCDF files**, enhancing ionosphere mapping resolution. Extracted insights from location-dependent electron density data from ISS FPMU to apply **3D data visualization**. Recognized as a **Global Finalist**. (October 2022)
- **Reddit Hate-Speech Detection**: Developed a **machine-learning model for hate speech detection** in Reddit posts. Used advanced **NLP** techniques like **word embeddings** to capture **semantic relationships**. Applied **SVMs & decision trees** for accurate classification. (October 2022)

## SKILLS SUMMARY

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- **Languages**: Python, R, MATLAB, SQL, Julia, Bash, C++.
- **Libraries**: ML/DL (PyTorch, TensorFlow, JAX, Scikit), Scientific (Pandas, Numpy, Scipy), API Development (FastAPI, Flask), Domain-Specific (Bioconductor, Qiskit, Astropy).
- **Typesetting Software**: L<sup>A</sup>T<sub>E</sub>X.
- **Platforms & Tools**: HPC (ARC, DRAC), Containers (Apptainer, Docker), LLM Serving (Ollama), Dev Tools (Shell, Git, VSCode, Jupyter, Anaconda), Cloud (AWS, GCP).
- **Lab Skills**: Experimental – Material growth (flux/solid-state), Neutron scattering, X-ray diffraction (XRD), EDS, PPMS/MPMS, DSC, TGA, Glovebox, Arc-melt furnaces, Analogue Circuits.
- **Communication**: Academic Writing, Public Speaking, Mentorship, Leadership, Event Management, Peer-counselling.

## TEACHING, MENTORSHIP, AND LEADERSHIP EXPERIENCE

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- **Undergraduate Teaching Assistant – Energy & Waves**  January 2021 – April 2023  
*Department of Physics & Astronomy, UBC* *Vancouver*
  - **Mentored student progress**: Supported **2000+ students** across multiple semesters of introductory physics, bridging knowledge gaps & facilitating the mastery of complex physics concepts.
  - **Facilitated active learning**: Leveraged strong **science communication skills** to translate difficult course material into accessible content, fostering an **engaging & supportive learning environment** during lectures & tutorials.
  - **Evaluated academic performance**: Invigilated exams & **graded assignments, midterms, & finals**, providing **constructive feedback** to support student development & academic integrity.
- **UBC Astronomy Club**  May 2023 – April 2025  
*Advisor & Outreach Coordinator* *Vancouver*
  - **Spearheaded outreach initiatives**: Coordinated & advised on the club's strategic direction, successfully fostering **engagement with diverse audiences** across the university & local community.

- **Synthesized complex scientific concepts:** Simplified advanced astronomy topics into **digestible content** for **social media reels & podcasts**, increasing accessibility for the general public.
- **Orchestrated educational lectures:** Organized & delivered interactive presentations to **school students**, bringing the wonders of space into **educational settings** to contribute to inspiring the next generation.

- **Residence Advisor**

📅 August 2021 – August 2023

*Student Housing & Community Services, UBC*

*Vancouver*

- **Enhanced residence life:** Facilitated a supportive living environment by utilizing a peer-to-peer approach to promote wellness, safety, & security for **1200+ students** in residence.
- **Established healthy communities:** Organized and led events & meetings to drive community building & personal development, **cultivating an atmosphere of understanding, inclusivity, & belonging.**
- **Community standards:** Communicated residence guidelines with **openness & assertiveness**, ensuring the well-being of residents, & fostering a **cooperative & supportive environment.**
- **Crisis management:** Successfully managed & resolved high-stress situations, including **medical emergencies, mental health crises, & conflict resolution.** Maintained **composure & professionalism** while providing compassionate support to residents.